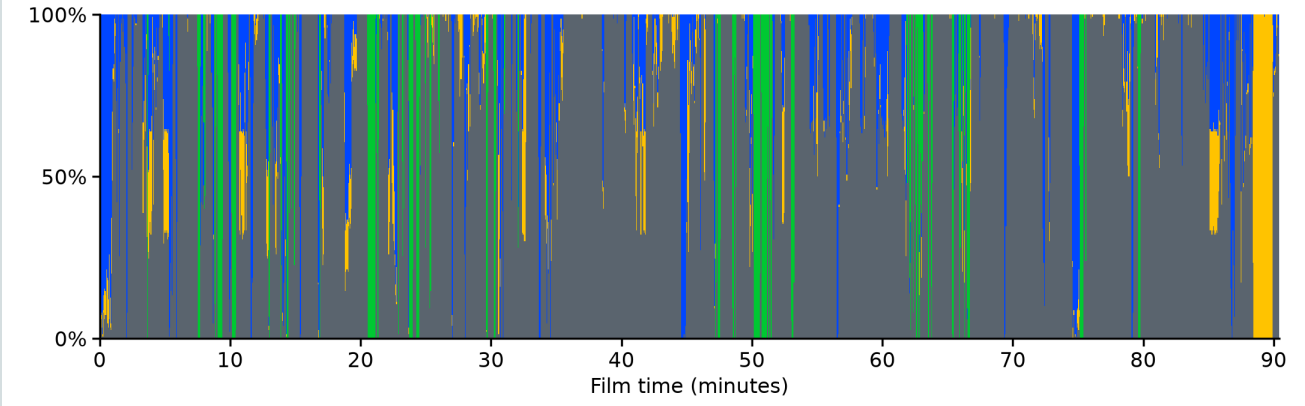
# The Champagne Spy, Restored

*A film that survived only on a DVD made for televisions that no longer exist, brought back the way it was meant to be seen: one laptop, an AI, and a rule to invent nothing the DVD didn't already hold.*

A film survives only in the formats it was released in, and formats age faster than the pictures inside them. A DVD pressed in 2009 was made for a kind of television that no longer exists. Restoration is the craft of recovering what such an original holds without adding anything to it, and it has mostly belonged to specialists with weeks to spend and budgets to match. This is that craft practised by one person and an AI on a single laptop, in two days, at a scale no human would sit through: all 271,174 half-pictures of one documentary (a DVD stores every frame as two) measured, judged on their own evidence and checked. The result looks the way the film was meant to look, and every change can be proved.

The only DVD of *The Champagne Spy* (2007), a German release from 2009, stores every frame of the film as two photographs taken 20 milliseconds apart and woven together line by line. That is interlacing: the way television was broadcast for most of the twentieth century, inherited by European DVDs made to the PAL television standard. A tube television drew the two halves one after the other and the seams never showed. A modern flat screen lights every pixel at once, so it shows both halves together and every seam between them. Much of the film is archive footage, old television and 1960s newsreel, and the DVD wears that history on every moving edge: comb-like stripes, shimmering street scenes, a German dub that drifts out of sync as the film runs, and on-screen titles an English viewer can't read. The restoration took the DVD apart to its smallest unit, the half-picture, worked out what each one was, and rebuilt the film from there. It now plays at 50 full frames a second, the dub sits on the lips to within a millisecond, and every German title has an English line.

> **Scope and rights.** The disc was purchased and carries no copy protection, so nothing was circumvented; the restored file exists for personal viewing only and is not distributed. What is shared is the method, the measurements and the code. Frames from the film appear here only as small crops for technical commentary; the film is © its producers and Neue Visionen.

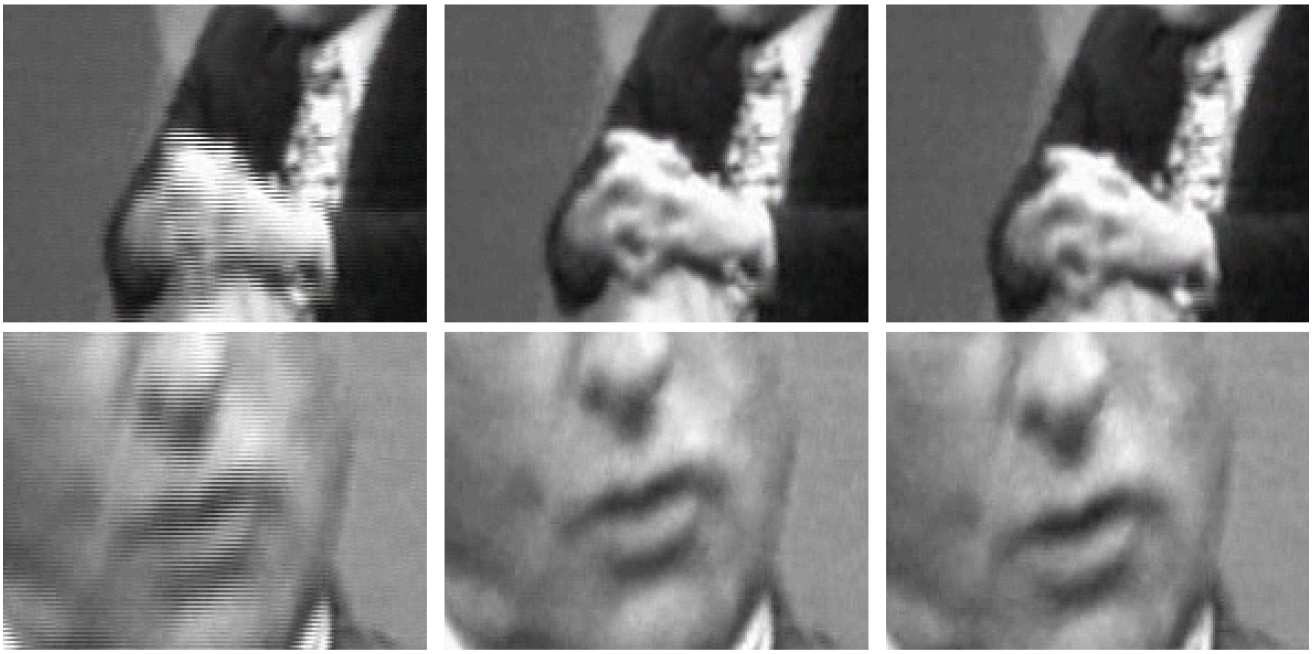



Every one of the film's 271,174 output frames, first minute on the left, last on the right, coloured by how it was made. **Slate:** the original DVD's own picture, joined as stored (74.5%). **Gold:** archive film the DVD had stored with its halves mismatched, repaired by moving its pixels back where they belonged (4.8%). **Blue:** true video motion, where the missing half had to be calculated (12.2%). **Green:** archive shots carrying old line noise, rebuilt clean (8.5%). The solid gold at the end is the rolling credits; the long slate stretches are the modern interviews.

A 7.92 GB DVD went in. The restoration produced a master that, stored without any loss, would fill about 110 GB, every half-picture accounted for and 85% of its pixels the DVD's own. That master was compressed back down to 6.2 GB for the film itself, roughly the space the DVD's own film streams occupy, while carrying twice the frames, finer colour steps and a German soundtrack stored without loss. Then every one of the 271,174 frames was scored against the master with VMAF, the picture-quality measure Netflix built to predict what viewers can see. The film averages 96 out of 100; the worst one per cent of frames stay above 91; at television size the two can't be told apart by eye. That was the goal, a better film than the DVD ever showed in a file no bigger than the DVD, with the measurements to prove it.

## Every frame on the original DVD holds two moments in time.

A PAL television camera never recorded whole frames. It captured 50 half-pictures a second, alternately the odd lines and the even lines, and the DVD stores them woven together in pairs, so anything that moved between the two captures splits into stripes on a modern screen. The usual fix is to blend each pair into one frame, which throws away half the motion and softens everything else. Here every half-picture became a full frame of its own: 135,587 DVD frames became 271,174, and motion the DVD had chopped into 25 steps a second plays at the 50 it was shot at.



Archive television, zoomed on a moving hand (74:34) and a talking mouth (0:44). Left: one frame of the original DVD, two moments woven together. Middle and right: the two clean frames that were inside it, 20 milliseconds apart.

## All 271,174 half-pictures were decided one at a time.

This film mixes four kinds of material, sometimes inside a single shot: film transferred as whole frames (progressive), video stored as two half-pictures (interlaced), still title cards, and archive footage transferred badly. No single treatment could be right for all of it, or even for each of its 908 shots, so the treatment was chosen for every half-picture on its own. Each was compared with its three possible partners: the other half of its own frame, and the nearest halves of the frames before and after. Wherever a partner had been captured at the same instant, the two were joined. That is the DVD's exact picture with no calculation at all, and it accounts for four frames in five. Only where no same-moment partner exists was anything calculated. The barcode above is the record of those decisions.

Deciding which partner, if any, was captured at the same instant takes evidence, and no single test could be trusted on its own. A standard comb detector, which looks for the stripes, also fires on clean hair, skin and fabric. A blurred comparison between half-pictures, which ignores grain, can't see fine combing but does reliably show which neighbour shares the instant: the right partner differs from the half-picture by less than half as much as the wrong one. The two were combined, plus a rule for still frames (their halves differ by a few hundredths of a brightness level, grainy film by one to three), so a title card is never mistaken for motion. Each kind of decision was then checked by rendering random samples three ways: the chosen pairing, the rejected pairing and a version calculated by the deinterlacer described below. The chosen pairing matched the calculation; the rejected one was a mess of stripes.

## Archive film the original DVD had scrambled was put back exactly.

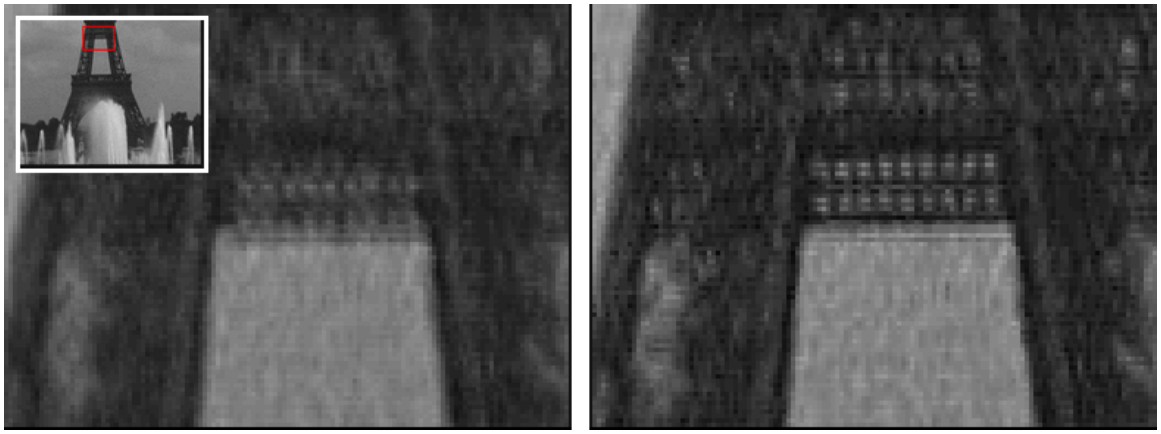
A good deal of the archive film reached the original DVD with its halves out of step: each stored frame joins the second half of one film frame to the first half of the next. Played as stored, the Paris street at 3:43 shimmers with fine lines on every car and fence post. The matching pass found the true partner of each of those halves one frame over and re-paired them. The pixels were moved back to where the film had them; none was invented or sharpened. That is the gold in the barcode, 4.8% of the film, restored to the original film frames exactly.



Archive film, 3:43, the fence zoomed. Left: as the original DVD stores it, two mismatched halves interleaved; the inset shows the whole frame and where the zoom is. Right: the same halves re-paired with their true partners.

## Where the picture was missing, it was calculated; the DVD's own lines were kept exactly.

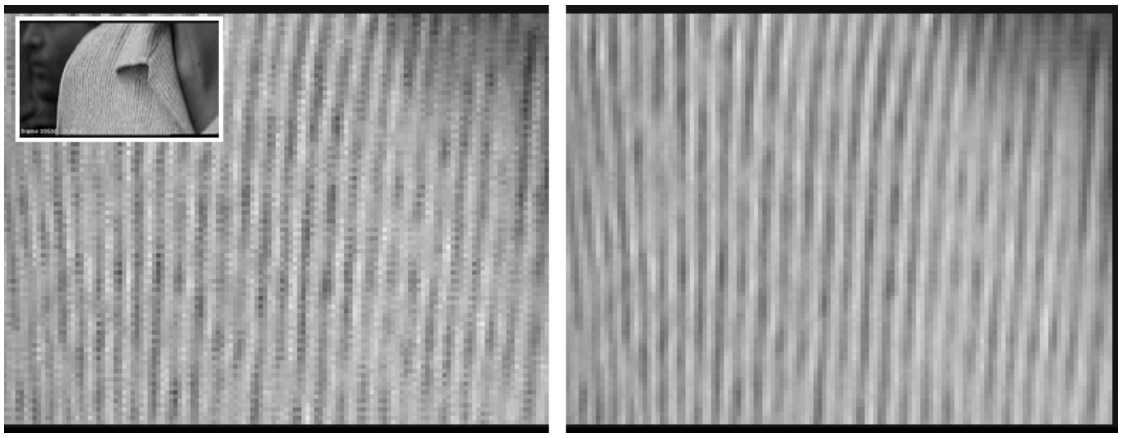
True video motion, the blue in the barcode, has no same-moment partner: the other half of that instant was never recorded. For those frames the missing lines were calculated with QTGMC, an open-source deinterlacer from the AviSynth and VapourSynth video-processing community, written by an author known as Vit and refined by that community since 2010. Its name comes from the method it grew out of, a temporal Gaussian motion-compensated filter, and that is a fair description: it tracks how things move through the neighbouring frames and uses that motion to work out what the absent lines should hold. It was run in a mode that keeps the DVD's own lines exactly as they are and calculates only the missing ones. The standard mode, which also reworks the kept lines, was tested and rejected: it shifted them by up to several brightness levels and softened fine detail such as the Eiffel Tower's ironwork. Measured afterwards, 88% of the pixels in the finished film are identical, bit for bit, to the original DVD.



The Eiffel Tower, zoomed to the first platform; the inset shows where. Left: the deinterlacer's standard mode, which reworks every line. Right: the mode used here, which keeps the DVD's lines and fills in only the missing ones; the ironwork stays crisp.

## Old line noise was cleaned only where measurement said it helped.

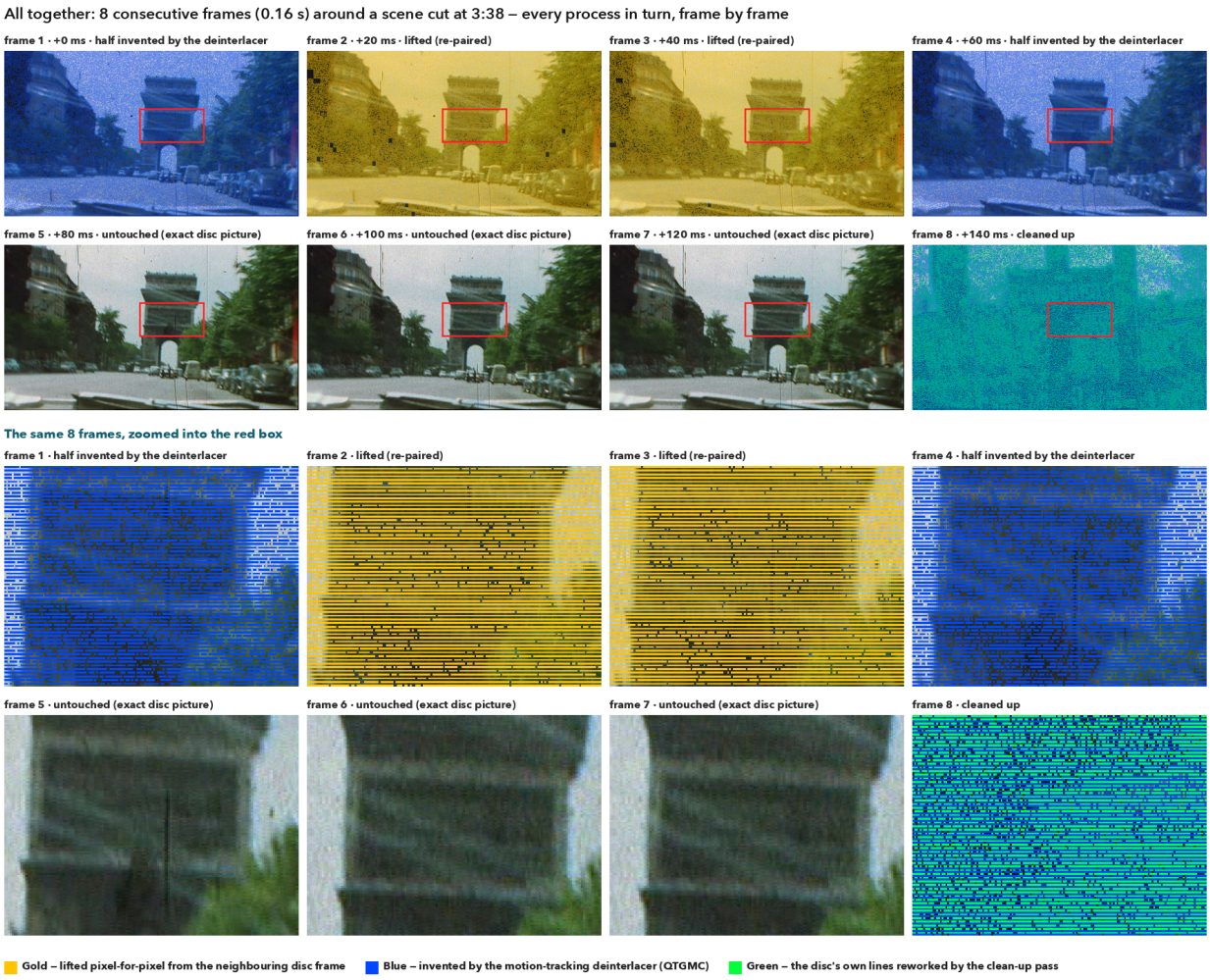
Some archive shots have a different problem: interlace noise baked into the old film-to-video transfer itself, a fine horizontal weave over the whole picture even where nothing moves. In those shots, keeping the DVD's lines means keeping the noise. Every one of the 908 shots was measured for it, and 116 of them, 7.7 minutes of the film, crossed the point where a full rebuild visibly won. Those got the deinterlacer's complete rebuild, which recomputes the DVD's own lines clean: the green in the barcode. The measured line noise in those shots fell from a median of 0.187 to 0.0001. Below that point the rebuild only softened detail, so the exact picture stayed, and it was never applied to text or credits.



23:46, a ribbed shirt the transfer had turned into a moiré (the rippling pattern two fine grids make when they overlap), zoomed three times; the inset shows it at television size. Left: the DVD's exact lines. Right: the clean rebuild.

## Four treatments, chosen 50 times a second.

Put together, the three processes and the untouched frames make a film in which the treatment can change every 20 milliseconds, and often does. Across one cut at 3:38, eight consecutive frames use all four: the deinterlacer where a half is missing, re-pairing where the film was scrambled, nothing at all where the DVD's picture was already right, and the clean rebuild on the first frame of the next shot. No frame mixes gold with the others. Each half-picture answers one question, whether its other half was photographed at the same instant, and the answer holds for the whole picture.




Eight consecutive frames, 20 milliseconds apart, across the cut at 3:38, tinted by the method that built each one; the bottom rows zoom into the red box. Blue: missing half calculated (frames 1 and 4). Gold: re-paired film (2 and 3). Untouched (5 to 7). Rebuilt clean (8, the next shot's first frame).

## One quality setting, chosen from 60 test encodes, then every frame checked.

The restored master is enormous; what plays is an H.265 file, the compression format modern televisions and phones play natively. Instead of a bitrate, the encoder was given a constant quality target, CRF 14. That setting came out of 60 test encodes, 30,000 frames in all, with every frame scored by two picture-quality meters and checked by eye at television size. With a quality target rather than a size, each scene takes the bits it needs. The result is 50 frames a second, 10-bit colour (finer steps between shades, so skies and shadows don't band) and 6.2 GB. Every one of the 271,174 output frames was then compared with the master using VMAF (Video Multimethod Assessment Fusion), the Netflix score mentioned at the start, developed with university labs and released as open source in 2016. It predicts, on a scale of 0 to 100, how a panel of viewers would rate a compressed frame against the original; it was trained on human ratings, so the number stands for what people see rather than for raw pixel error. The film averages 96.1, and the worst one per cent of frames still score 91.

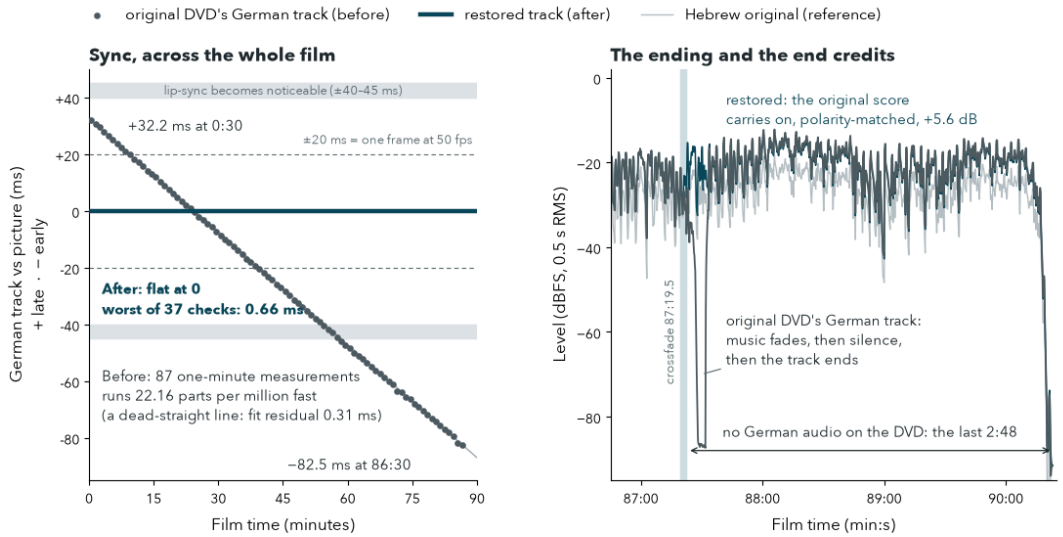



The Eiffel Tower's ironwork, zoomed. Left: the restored master. Middle: the delivered file (CRF 14). Right: a CRF 18 test encode, which would have been about 35% smaller; it flattens the film grain slightly and was rejected. At television size the delivered file and the master can't be told apart by eye.

## The German dub was re-timed to within a millisecond.

The original DVD's German track doesn't fit the picture. It runs 22 parts per million fast, so it starts 32 milliseconds late and ends 83 early, enough to see on the lips. Measured against the original Hebrew track, which is locked to the picture, at 87 one-minute windows, the drift turned out to be a straight line, to within a third of a millisecond, and the track was re-timed against it. The worst remaining error, across 37 verification windows, is 0.66 milliseconds, a thirtieth of a frame.

The DVD's German track also fades its music out just before the end credits and never comes back. The original score was carried through instead, matched in loudness and in waveform polarity (the German mix is inverted relative to the original, so it was flipped to match), with a three-second crossfade at 87:19.5.




Left: the German track's offset against the picture, one measurement a minute; the dots are the original DVD, the flat line is after re-timing. Right: sound level through the ending. The DVD's German track fades and dies before the credits; the restored track carries the original score through at matched level.

## Every German title on screen now has an English line, timed to the title itself.

The original DVD stores its English subtitles as pictures. They were read by OCR (optical character recognition) into text, 788 lines, each one corrected, so they draw cleanly at any size. But the film also puts German on screen that the DVD never subtitled: the title card, the opening credits, the name captions, the epilogue cards. Fifteen English lines were written for those, and each was timed by measuring the German title's own brightness frame by frame, so the line appears as the title fades in and leaves before the title has fully faded.

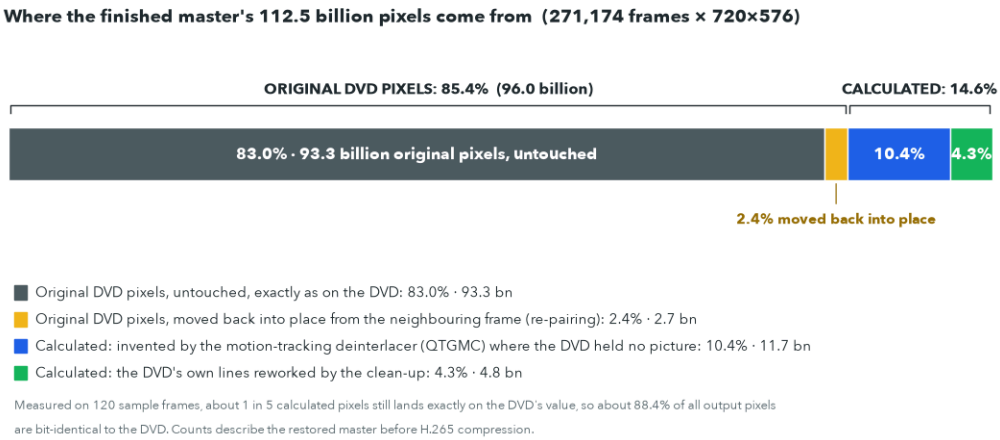



An opening credit measured frame by frame. Its English line runs from 92.80 s to 94.80 s, the last frame on which the title is visible.

All of it ran on one laptop: an evening of analysis, then eight hours of encoding overnight. Before a single final frame was encoded, the whole film had been measured eight times over and rendered in full twice, 542,348 frames, just to find the 116 shots where the clean-up helped. The two interviews on the DVD were treated the same way as the film, and two widescreen trailers that the DVD had stored with black bars inside an old 4:3 frame were reframed to fill the screen. Nothing was sharpened, denoised or resized.

## A closing thought.

A lot of what is sold as video restoration invents: an upscaler or an AI enhancer adds detail, and the result can look impressive while having less and less to do with what was photographed. The rule here was to invent nothing the original DVD didn't hold, and to prove it. Six of every seven pixels on screen are the DVD's own, yet the film moves at twice the rate, the stripes are gone, the dub is on the lips and the titles can be read, because the DVD was already carrying nearly all of this. It only needed every half-picture asked the same question, and answered one at a time.




Where the finished master's pixels come from: six of every seven are the original DVD's own, and 88.4% of all of them are bit-identical to it.

A DVD is usually treated as a finished object: you play it, and what you see is what there is. Treated as data instead, 271,174 half-pictures with a measurable relationship to each other, it turns out to contain a better film than it ever showed.

> **See it move.** Print cannot show motion, and motion is where the difference is biggest. The same scenes as short clips, the original DVD on the left and the restoration on the right, plus the German ending with sound, are in the `media/` folder of the project repository alongside this document.

> Frames reproduced in this document are from *The Champagne Spy* (2007, dir. Nadav Schirman), © its producers and Neue Visionen, shown as small crops for technical commentary. Restoration, measurement and text: Levi Horsley.